

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

**COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems**
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. **Intelligent Remote Patient Monitoring Using Wearable IoT Devices** (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Summary

Digital parameters concerning a person's health status can be continuously measured by wearable technology and shown in real time. Photoplethysmography-based heart rate monitors and hardware and software activity monitors are just two examples of the many readily available devices seen as a significant part of the fitness market and modern lifestyle. The reality that so few wearables are permitted in the healthcare system presents a considerable barrier to their medical usage. The four most crucial metrics in the healthcare system—heart rate, temperature, heart rate, and breathing rate—are frequently only taken

during a physical examination by a primary care doctor. Continuous measurements are possible but not very portable; nonetheless, wearable technology makes it possible to measure these devices longitudinally and accurately. The main idea behind wearable technology in the healthcare system is to provide doctors and patients with high-quality, clinically relevant data. Reviews biosignal measurements, including heart rate, temperature, pulse rate, breathing rate, physical activity, sweat, and emotion. Society has greatly benefited from the rising usage of mobile technology and innovative gadgets in health. The advantages these technologies offer are progressively embraced by health professionals, leading to a notable improvement in both clinical settings and outside of them in terms of medical care. With the strain on healthcare infrastructure and healthcare workers due to the ongoing COVID'19 pandemic, the need for novel ways to simplify the interaction between patients and physicians has increased. The aim is mainly to reduce face-to-face interactions and free more time and resources for those who urgently need it. This paper presents a Telemedicine system which can be used as an alternative method to a doctor's visit. The proposed system is considered as an interface that remotely connects the patient and the doctor. The system regularly measures and uploads readings from the patient to a database which the doctor can remotely review to decide on the state of the patient. The whole system is based on the internet of medical things (IoMT). In the proposed system, sensors collect and send data over the internet using the WIFI module connected to a node MCU controller. The developed prototype uses two sensors, one is used to measure both the percentage of oxygen in the blood and heart rate while the other sensor measures the temperature of the human body. The aim of this work is to help patients, by offering them this solution which can help them get the care they require from the comfort of their home and at the same time help doctors remotely give them the suitable treatment while reducing the need for unnecessary face-to-face interactions.

Wearable Devices for Patient Monitoring System using IoT

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Abstract—The most crucial components of life is health. People could still not access adequate healthcare, nevertheless. It is brought on by restrictions on the technology used in hospitals and on how people may travel to hospitals. One of today's most popular topics, the Internet of Things (IoT), already provides many solutions, including in the healthcare industry. The field of wearable medical technology has emerged as one of the promising healthcare technologies. By actively collecting physiological indicators and monitoring metabolic status, these intelligent sensors help individuals lead healthier lifestyle and provide continuous healthcare

healthcare systems are unavoidable due to the effective data processing in numerous devices, including smart watches, smart clothing, and medical wearables, as well as the consumer-oriented services of IoT technology [3]. Figure 1 shows wearable technology.

